



LiDAR and Camera-based Navigation Algorithms for Lane-keeping in Unstructured Roads

BTP Poster

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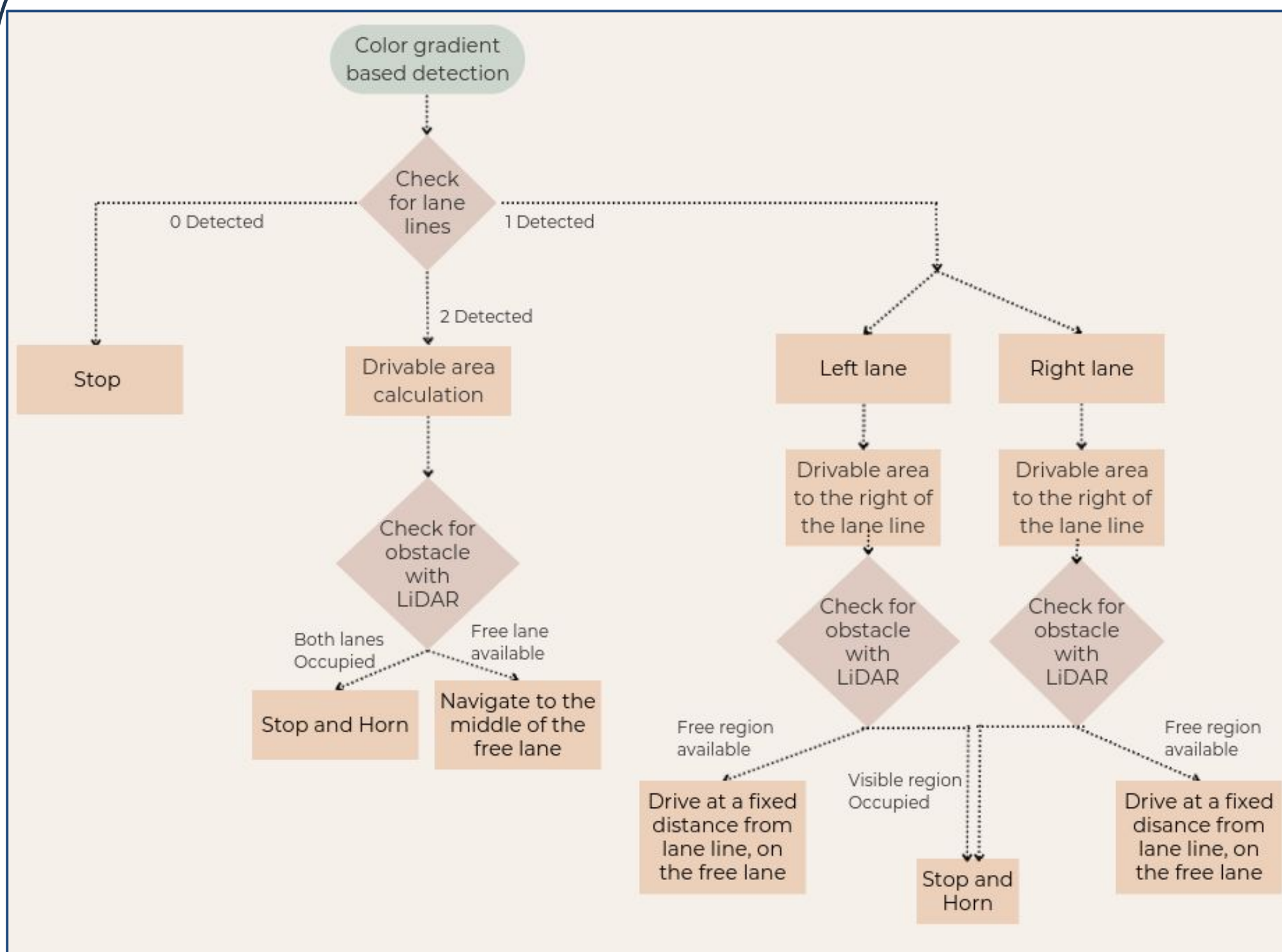
Abstract

This research presents a LiDAR-camera rule-based approach for lane-keeping in unstructured environments using the CARLA simulator. Using traditional computer vision methods, such as the Line Segment Detector (LSD), offers real-time performance suitable for resource-limited systems. In contrast with deep learning models, this method guarantees explainability and low computational overhead. The system is tested under diverse weather and lighting conditions, with reliable lane detection and adaptability to unstructured terrains, and is thus a lightweight benchmark for future autonomous driving applications.

1. Introduction

Autonomous driving is a revolution in transportation that depends on advanced sensors, real-time data processing, and artificial intelligence to enable vehicles to navigate the roads with minimal or no human intervention. It can transform urban mobility, logistics, and personal transport by enhancing safety, reducing traffic congestion, and expanding access to transport. Autonomous vehicles need to be capable of perceiving and understanding rich environments, anticipating and reacting to invisible threats, and making real-time high-stakes decisions independently. The target is to move beyond the constraints of today's lane-keeping systems and unlock the complementary capabilities of Camera and LiDAR to deliver superior real-world autonomous driving. Simulators such as CARLA provide the perfect environment for this, with the capability to personalize cityscapes, sensor setups, and the capability to simulate various weather and lighting conditions.

2. Methodology



The process starts with color gradient-based detection of lanes followed by a decision tree based on the number of lane lines detected. If lanes are not detected, the car stops for safety. If two lanes are detected, it computes the drivable region between them. When a single lane is detected, the system identifies whether it's left or right lane marker and computes drivable region accordingly. The decision logic then diverges based on obstacle presence

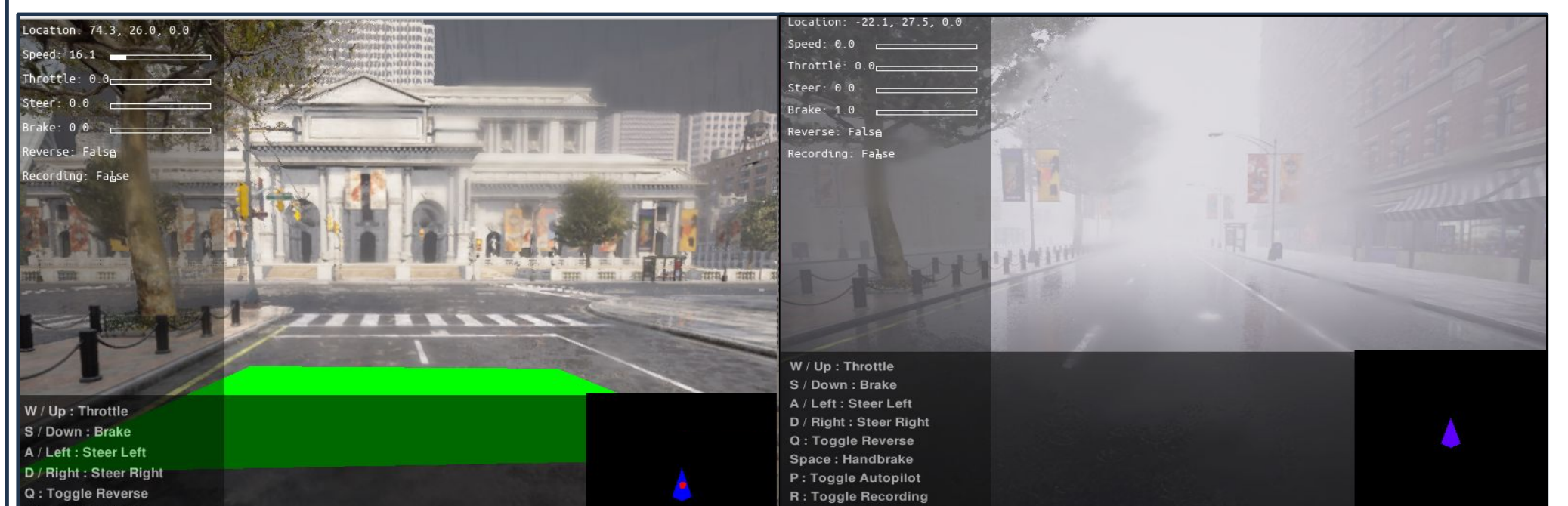
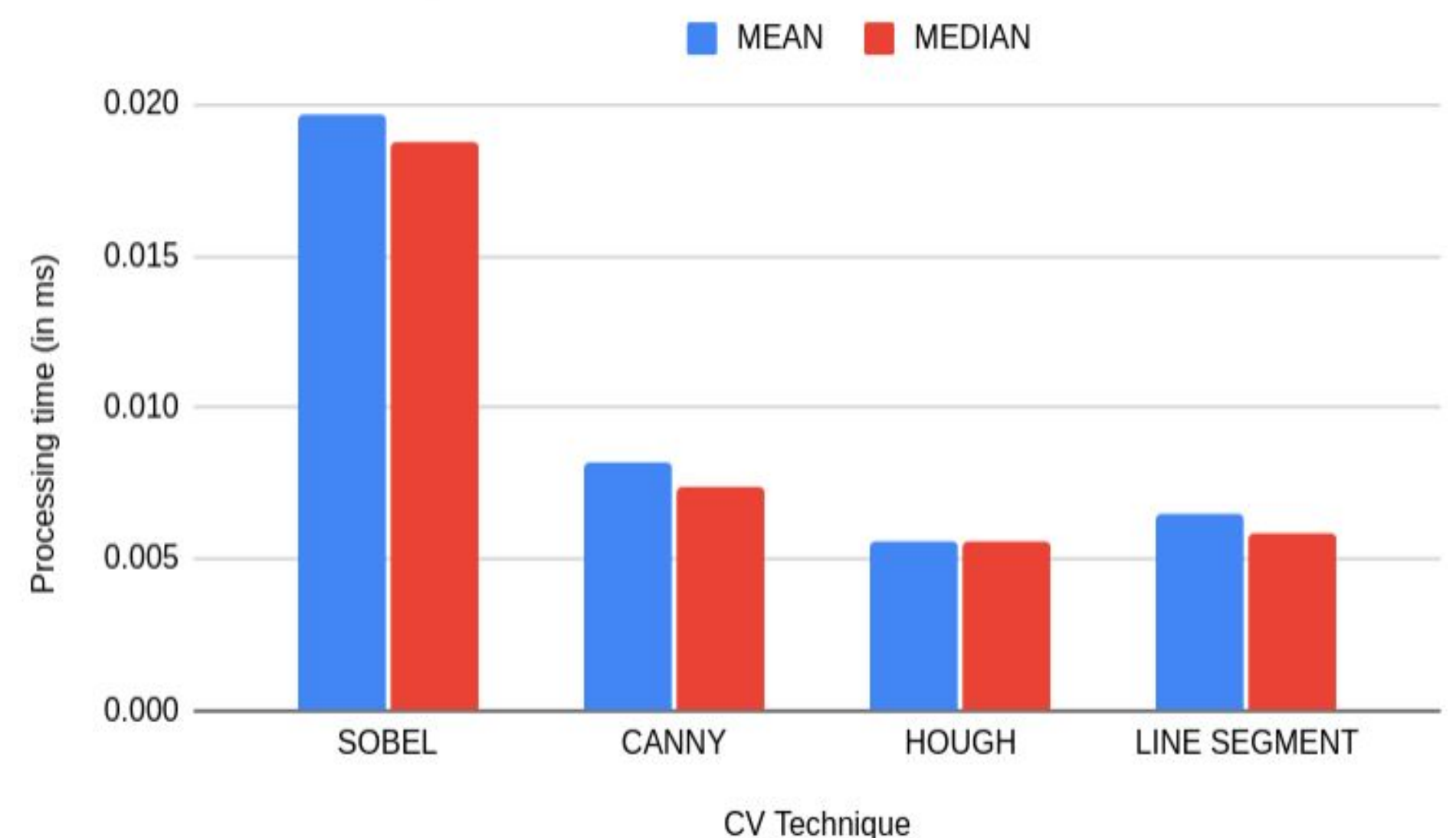
0: If both lanes are occupied, the vehicle stops and activates the horn

1: For single-lane scenarios, it maintains a fixed distance from the detected lane line when the region is clear. If the lane beside is available, the vehicle would take that, or stops if obstacles are present in the visible region.

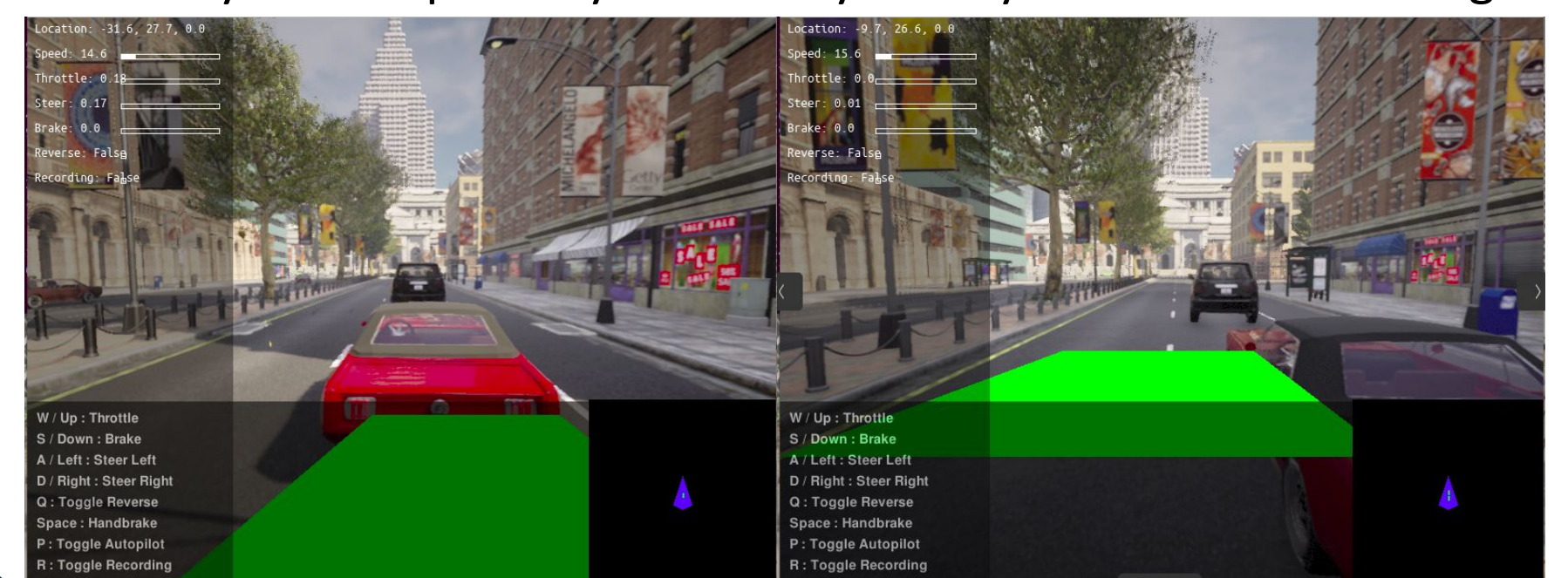
2: If a free lane exists and that both lanes are visible, it navigates to the center of the closest lane and continues moving along that.

3. Results & Discussion

Performance comparison



The proposed system follows a hybrid approach and achieves robust lane detection accuracy in unstructured scenarios, outperforming camera-only systems while maintaining minimal lateral deviation from inferred paths. The system adapts dynamically to challenges such as rain, uneven terrain, and curved lanes. While current performance demonstrates reliability in tested conditions, future iterations would address edge cases such as dense fog, where sensor limitations reduce visibility. Enhancing perception in low-visibility environments along with road sign interpretation rules will further solidify the adaptability and safety in fully unstructured settings.



4. Conclusion

The study of conventional computer vision methods for lane detection indicates significant trade-offs between computational complexity and accuracy. The Line Segment Detector (LSD) approach demonstrates better performance through an optimal trade-off. The proposed algorithm combining the lane information with LiDAR showed resilience under varying lighting and weather conditions, reinforced the robustness. Despite being rule-based, it sets a reliable baseline for future autonomous navigation research.

References

- [1] <https://github.com/carla-simulator/carla>
- [2] <https://github.com/erdos-project/pylot>
- [3] <https://github.com/PJLab-ADG/PCSim>

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